NATIONAL INSTITUTE OF TECHNOLOGY WARANGAL
(An Institute of National Importance)
Department of Computer Science & Engineering

CS428 Design of Secure Protocols

Autumn Semester, 2026

1. Course Contents:

Topic	No. of Lectures
Formal Models for Symmetric Key Encryption and Integrity	**20**
Introduction to SSL protocol, TLS Layer protocol, Cipher suits Session and Connection establishment between client & server, Overview of Types of Attacks: Ciphertext only attack, Chosen Ciphertext attack, Chosen Plaintext attack, Known plaintext attack,	3
Definitions & overview: Indistinguishability, Perfect secrecy, one time pad, Shannon theorem, Computational Security, Hardness assumption: Factorization, Discrete logarithm problem, Pseudorandom Generators and Stream Ciphers & Block Cipher	3
Secure Fixed-Length Encryption Scheme, Security for Multiple Encryptions	2
Chosen-Plaintext Attacks and CPA-Security, CPA-Secure Encryption from Pseudorandom Functions	3
Modes of Operation	1
Chosen-Ciphertext Attacks	1
Fixed-Length MAC, Domain Extension, CBC-MAC,	2
Authenticated Encryption	2
CCA-Secure Encryption	1
The Merkle–Damgard Transform, Message Authentication Using Hash Functions, The Random- Oracle Model	2
Formal Models for Asymmetric Key Encryption and signatures	**12**
Security against Chosen-Plaintext Attacks	2
Security against Chosen-Ciphertext Attacks	2
CDH/DDH-Based Encryption, El Gamal Encryption	2
DDH-Based Key Encapsulation, CDH-Based KEM in the Random-Oracle Model	2
Chosen-Ciphertext Security	1
Digital Signature Schemes, Zero knowledge proofs	3

Protocols and System	**8**
Secret Sharing and Threshold Encryption	2
Secure Authenticated Key exchange protocols	2
Secure Group Key agreement protocols	2
Application area : Electronic Voting, Vehicular Network, Healthcare	2

2. **Books & References: Properly Formatted along with listing of possible internet sources.**

Text Book

Introduction to Modern Cryptography	Jonathan Katz and Yahuda Lindell, "", CRC press, 2008
Foundations of Cryptography	Oded Goldreich, ", Vol. I and II", Cambridge University Press, 2007.
Cryptography & Network Security	Behrouz A. Forouzan and Debdeep Mukhopadhyay, Mc Graw Hill publication
Reference Books	
Handbook of Applied Cryptography	Van Oorschot, Paul Scott, A Vanstone, A J Menezes, CRC Press, 2004